

Stochastic Diffusion Search in Skin Cancer Detection ¹ $\LaTeX$

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Abstract. Medical detection field is subject to high dimensionality problems, one of such is skin cancer detection which have [thousands] of features, this paper presents a novel approach in utilisation of a swarm intelligence algorithm Stochastic Diffusion Search in detecting skin malignancy from smartphone taken images.

1 Introduction

Skin cancer spread and urgency Skin cancer accounts for one third of all cancers worldwide, moreover, it generates the most urgent referrals of any cancer. In the UK, the incidence of skin cancer is on an upward trajectory [2] , leading to an increase in referral rates for assessment by over 10 percent year on year. Despite sustained and considerable effort from dermatology departments, this has led to wait-lists growing significantly. In 2024 ,100,000 patients waited more than 28 days after urgent cancer referral for their diagnosis [1]. Early detection greatly improves the prognosis of patients with malignant melanoma. Skin cancer detection is difficult because of differences in lesion size, shape, and lighting condition, studies have reported Artificial intelligence systems' accuracy matching or exceeding dermatologists for the diagnosis of skin cancer from clinical and dermoscopic images. SDS-based Feature selection is especially competitive when feature count larger than 10. [5] [9] [10]

2 Background

The Skin The skin is the largest body organ. It is composed of two primary layers: the outer layer, known as the epidermis, primarily consisting of keratinocytes, and the dermis which comprises fibroblasts, mast cells, macrophages, and Langerhans cells. Melanocytes reside within the basal layer of the epidermis and play a crucial role in supplying and transferring melanin to keratinocytes, resulting in skin pigmentation and protection against UV damage.

skin cancer is a heterogeneous group of cancers and a world-wide epidemic. It is categorized into two major types: melanoma skin cancers (MSCs), which develop from skin pigment cells called melanocytes and non-melanoma skin cancers (NMSCs) which develop from keratinocytes . and These cancers account for 90 percent of all skin cancers.[6] [7] melanoma skin cancers (MSCs), which develops from skin pigment cells called melanocytes. Skin cancer detection methods: The ABCD checklist has been reported to have a sensitivity of 92 to 100 percent and a specificity of 98 percent for the detection of melanoma. [6]

Skin cancer early detection and treatment: Addressing skin cancer before they escalate into serious health problems not only enhances patient outcomes but also drives down healthcare costs, by reducing unnecessary referrals and ensuring that specialists can focus on patients that are most at risk. having a diagnosis app at hand fosters

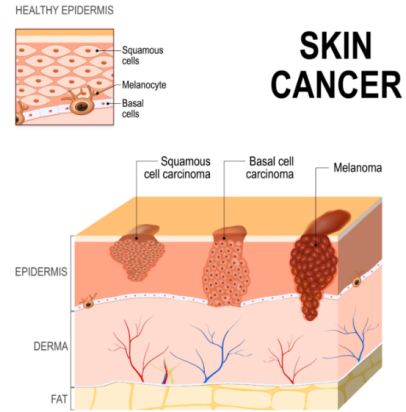


Figure 1.

Checklist item	Reassuring elements	Nonreassuring elements
A = Asymmetry Suggestive of melanoma if the lesion is bisected and the halves are not identical	Symmetric (benign)	Asymmetric (malignant melanoma)
B = Border irregularity Suggestive of melanoma if the border is uneven or ragged	Borders are even (benign)	Borders are irregular (malignant melanoma)
C = Color variation Suggestive of melanoma if there is more than one shade of pigment	One shade/even color (benign)	Two or more shades/uneven color (malignant melanoma)
D = Diameter Suggestive of melanoma if the diameter is greater than 6 mm	Diameter < 6 mm (benign)	Diameter > 6 mm (malignant melanoma)

note: Biopsy should be strongly considered if one or more nonreassuring elements are present.

Figure 2.

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community awareness regarding skin health issues, encouraging a culture of preventative care.

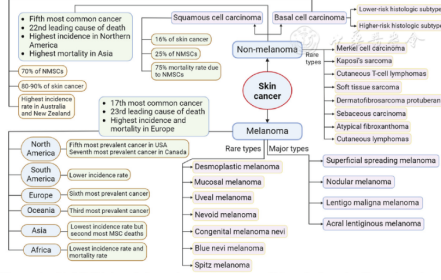


Figure 3.

Stochastic Diffusion Search algorithm (SDS) was proposed in 1989 as a population-based pattern-matching algorithm, it is a swarm intelligence algorithm which uses a guided random search technique, it introduces population of agents where each agent poses a hypothesis about the possible solution and evaluates it partially. successful agents repeatedly test their hypothesis while recruiting unsuccessful agents by direct communication. This creates a positive feedback mechanism ensuring rapid convergence of agents onto promising solutions in the space of all solutions. Regions of the solution space labelled by the presence of agent clusters can be interpreted as good candidate solutions. A global solution is thus constructed from the interaction of many simple, locally operating agents forming the largest cluster.[4] [3]

A sample reference can be seen here [?].

3 Experiments and results

The SDS algorithm operates by iteratively alternating between two phases: Test and Diffusion Initially, a population of N agents is created, each assigned a random hypothesis from the search space. In the Test phase, every agent performs a partial evaluation (a micro test) on its current hypothesis to gauge its merit. This is an inexpensive, randomized test that checks one aspect or component of the hypothesis rather than evaluating it fully. The outcome of the micro test is binary (success/failure); if the partial test passes, the agent is marked active, otherwise inactive. In other words, SDS “checks whether the agent’s hypothesis is successful or not by performing a hypothesis evaluation which returns a boolean value. Active agents are those whose hypotheses showed evidence of quality in the sample test, whereas inactive agents did not.

Following testing, the algorithm enters the Diffusion phase, wherein information is spread by agents with successful hypotheses. In the standard passive recruitment strategy used in SDS, only inactive agents initiate recruitment: each inactive agent randomly selects another agent from the swarm. If the chosen peer is active (i.e. holds a promising hypothesis), the inactive agent adopts the peer’s hypothesis. This one-to-one communication allows high-quality candidate solutions to propagate through the population. If the chosen peer is also inactive, then the recruiting agent discards its old hypothesis and instead generates a new random hypothesis to explore. Meanwhile, active agents typically retain their current hypothesis into the next iteration. By the end of the diffusion phase, some previously inactive agents will have gained better hypotheses from their active peers, and others will randomize, ensuring continuous exploration. The iteration then repeats with a new test phase on the updated population. This

cyclic process continues until a stopping criterion is met (e.g. a maximum number of iterations or convergence when enough agents become active and hold identical hypotheses). Ultimately, SDS can return the hypothesis supported by the largest number of active agents as the best solution, effectively leveraging the emergent consensus of the swarm.

The efficiency of SDS has been demonstrated in a variety of experimental settings, often showing fast convergence and robust performance in noisy search landscapes. One notable application is in the domain of medical image analysis. Al-Rifaie et al. adapted SDS to assist in detecting bone metastases from nuclear medicine scans (bone scintigraphy). In this approach, thousands of agents were deployed over a bone scan image, each agent hypothesizing a region of the image that might contain metastatic activity. The partial evaluation in this context involved an agent randomly sampling a small region (or pixel) within its hypothesized area to check for abnormally high radionuclide uptake, a hallmark of metastatic lesions. If the sampled signal intensity exceeded a threshold (indicating potential tumour activity), the agent became active; otherwise, it became inactive. Through many iterations of test and diffusion, the SDS swarm gradually concentrated its agents on areas of the scan with suspicious high-intensity signals. The result was an automated highlighting of likely metastasis locations. In their experiments on real patient bone scans whose diagnoses were confirmed by experts, this SDS-based technique could successfully identify regions of metastatic cancer on the images. The novel application of SDS to bone scan analysis yielded promising results, with the algorithm exhibiting statistically significant differences in agent activation patterns when processing healthy scans versus scans with metastatic disease. This indicates that SDS was able to effectively distinguish diseased bone images by the collective behavior of its agents. Such findings demonstrate the potential of SDS as a decision-support tool in medical imaging, complementing radiologists by automatically detecting abnormal patterns on the scans.[1] [2]

4 Discussion

Partial Evaluation and Fitness: A distinctive feature of SDS is its use of partial stochastic evaluation in lieu of full fitness computation. This can be formalized as follows. Let $h_i(t)$ denote the hypothesis held by agent i at iteration t . We define a fitness function $f(h)$ that represents the probability that a random micro test on hypothesis h will succeed (i.e. the hypothesis matches the target on a randomly chosen component). **Dynamics of Diffusion:** The diffusion phase then acts as a selection mechanism amplifying good hypotheses in the population. If an agent is inactive ($a_i = 0$), it seeks information from a random peer j . In standard SDS, this results in the update rule:

$$\vec{x}_i^{t+1} = \begin{cases} \vec{x}_j^t + \delta, & \text{if } a_i^t = 0 \text{ and } a_j^t = 1 \\ \text{rand}(\mathcal{S}), & \text{if } a_i^t = 0 \text{ and } a_j^t = 0 \\ \vec{x}_i^t, & \text{if } a_i^t = 1 \end{cases} \quad (1)$$

which encapsulates the passive recruitment strategy. Thus, an inactive agent will copy the hypothesis of a randomly chosen active agent (if it finds one), effectively spreading that promising hypothesis, or else assign itself a new random hypothesis to continue exploring. Active agents keep their successful hypotheses. Through this process, clusters of agents sharing identical hypotheses form and grow around high-fitness solutions. Theoretical analyses of SDS have shown that this simple mechanism guarantees, under reasonable conditions, convergence towards globally optimal solutions with high probability.

Moreover, the algorithm's complexity per iteration is linear in the number of agents and hypotheses tested, since each agent performs only a constant-time partial evaluation and a constant-time communication update. This leads to efficient scalability even for large search spaces. Studies have also highlighted SDS's robustness to noise and partial information, it can tolerate noisy fitness signals and still reliably home in on good solutions. One practical consideration is that SDS requires the problem to be decomposable into suitable micro tests (partial evaluations); designing an appropriate partial evaluation function $f(h)$ is key to applying SDS successfully. When such a function exists, SDS can be a powerful solver, balancing exploration and exploitation without the need for intricate parameter tuning. [8]

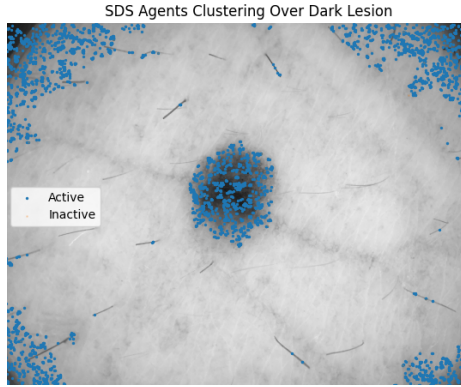


Figure 4.

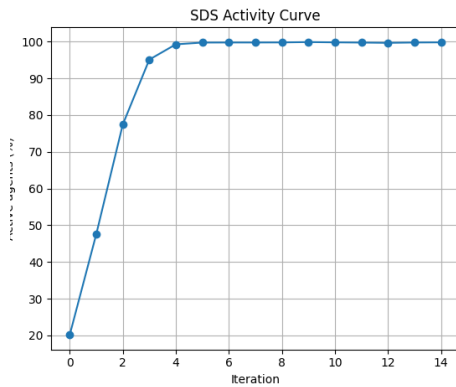


Figure 5.

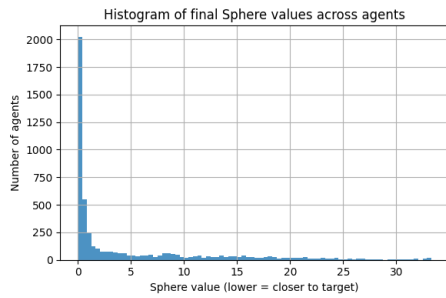


Figure 6.

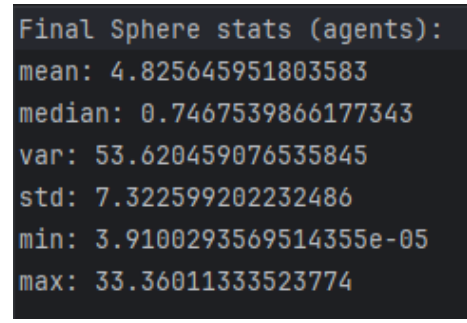


Figure 7.

5 Conclusion

In conclusion, early detection of skin cancer not only improves individual health outcomes but also offers significant societal and economic benefits by reducing unnecessary referrals and easing the burden on healthcare systems. The integration of Stochastic Diffusion Search (SDS) into diagnostic tools provides a robust, efficient, and decentralized method for identifying malignancies using image data. SDS enables agents to self-organize and converge on likely lesion regions through simple partial evaluations and peer-to-peer communication. Its effectiveness has been demonstrated in domains such as bone metastasis detection and medical imaging more broadly. However, ethical considerations such as the consequences of false positives or false negatives must be addressed when deploying such technologies in clinical settings. Future work should focus on optimizing image processing parameters, fine-tuning SDS for metastasis detection, and exploring hybrid models that combine SDS with classifiers like support vector machines or deep learning. Expanding to other swarm intelligence methods can further improve diagnostic accuracy across diverse skin cancer subtypes.

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